

LLM LAB ASSIGNMENT – 1

Task 1: Binary Classification using KNN Classifier

1. Abstract

This experiment implements a binary classification system using K-Nearest Neighbors (KNN) algorithm from scratch, without using a ready made KNN classifier from scikit-learn. The Breast Cancer Wisconsin dataset was used, containing 569 samples and 33 features. The data was divided into 80% training and 20% testing sets using a stratified split. The input features were also standardized from the training data. Five values of K (3, 4, 9, 20, 47) and five distance metrics (Euclidean, Manhattan, Minkowski, Cosine Similarity and Hamming) were evaluated. The best result was obtained using Euclidean distance with $k = 9$, achieving a test accuracy of 98.23%. The respective precision was 100% and recall was 95.24%.

2. Objectives

- Implement the K-Nearest Neighbors classifier from scratch.
- Use and 80:20 stratified training/testing split.
- Evaluate $K = 3, 4, 9, 20$ and 47
- Compare Euclidean, Manhattan, Minkowski, Cosine Similarity and Hamming Distance
- Select the best K and distance metric and testing accuracy.
- Calculate the confusion matrix, precision and recall for the best model.

3. Dataset Description

The experiment uses the Breast Cancer Wisconsin dataset. The dataset contains 569 samples and 33 features.

Property	Value
Total samples	569
Input features	30 numerical features
Benign samples (B)	357
Malignant samples (M)	212
Target encoding	B = 0, M = 1
Training samples	456
Testing samples	113

4. Data Preprocessing

The identifier column was removed because it does not represent any medical feature. The target diagnosis was converted into numerical labels, with B mapped to 0 and M mapped to 1.

The features have very different numeric ranges such as area and perimeter values are much larger than smoothness values. Therefore, standardization was applied using the training-set mean and standard deviation.

5. Train-Test Split

An 80:20 stratified split was implemented manually. Stratification maintains representation of both classes in the training and testing subsets.

Subset	Total	Benign	Malignant
Training	456	286	170
Testing	113	71	42

6. Distance Metrics

Different distance metrics are used to evaluate the results of the KNN classifier.

Metric	Description
Euclidean	L2 distance; $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Manhattan	L1 distance; $d = x_1 - x_2 + y_1 - y_2 $
Minkowski	General p-norm; p = 3 was used in this experiment. $D(x, y) = \left(\sum_{i=1}^n x_i - y_i ^p \right)^{\frac{1}{p}}$
Cosine	Uses 1 - cosine similarity, measuring angular rather than direct magnitude difference.
Hamming	Counts coordinates that are unequal, naturally more suitable for binary/categorical data.

7. Experimental Setup

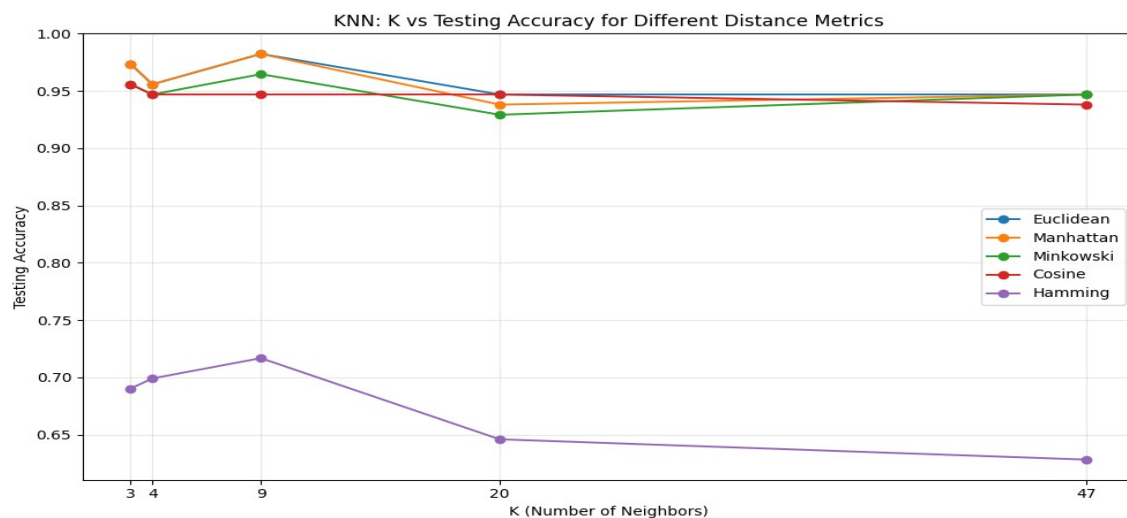
Parameter	Setting
Algorithm	K-Nearest Neighbors from scratch
Train/Test split	80% / 20%
Random seed	42
K values	3, 4, 9, 20, 47
Distance metrics	Euclidean, Manhattan, Minkowski, Cosine, Hamming
Selection criterion	Highest test accuracy

8. Experimental Results

All 25 combination of the five K-values and five distance metrics were evaluated. The following table shows the complete accuracy results in percentage form.

Metric	K=3	K=4	K=9	K=20	K=47
Euclidean	97.35%	95.58%	98.23%	94.69%	94.69%
Manhattan	97.35%	95.58%	98.23%	93.81%	94.69%
Minkowski (p=3)	95.58%	94.69%	96.46%	92.92%	94.69%
Cosine	95.58%	94.69%	94.69%	94.69%	93.81%
Hamming	69.03%	69.91%	71.68%	64.60%	62.83%

The highest observed accuracy is 98.23% achieved by both Euclidean and Manhattan distance when K = 9. The notebook selected the Euclidian with K = 9 as the best configuration because it appears first among the maximum accuracy rows after sorting the results.



9. Best Model

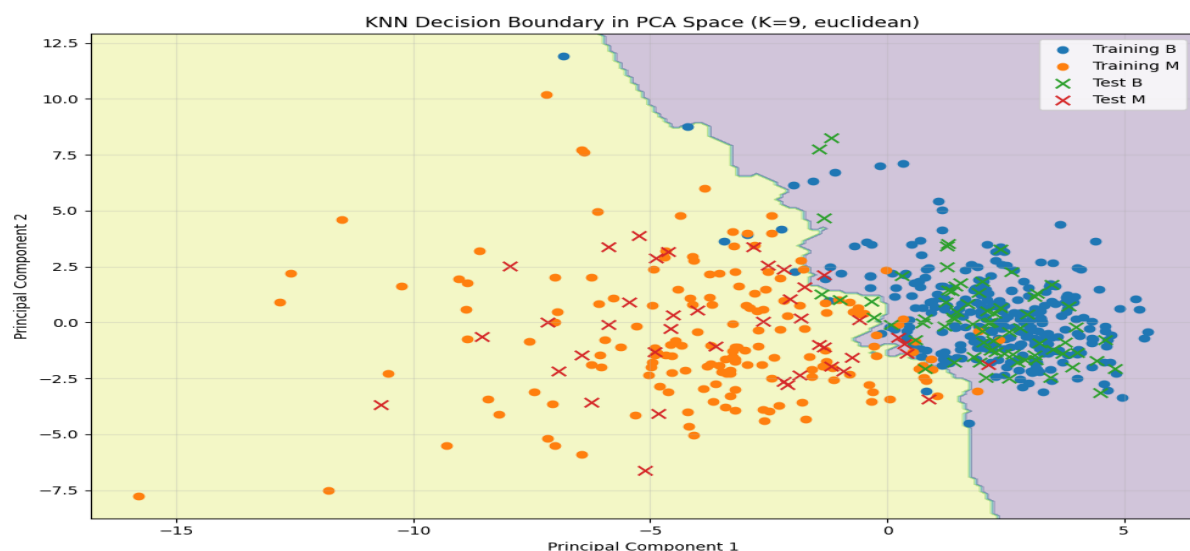
Measure	Best Result
Distance metric	Euclidean
K	9
Test accuracy	98.23%
Precision	100.00%
Recall	95.24%

10. Confusion Matrix

	Predicted B	Predicted M
Actual B	71	0
Actual M	2	40

11. Bonus: Decision Boundary

We plotted the Decision Boundary using the best model i.e., Euclidean Distance with $K = 9$.



12. Final Observations

- **Feature scaling is necessary** because the 30 features have very different numerical ranges.
- KNN is a **lazy learning algorithm**: it stores the training samples and performs distance calculations at prediction time.
- The best K and metric must be selected from the **test accuracies obtained in the experiment**, not assumed beforehand.
- Small K values are sensitive to individual training samples, while larger K values produce smoother decisions but can underfit.
- Euclidean, Manhattan and Minkowski measure different notions of geometric closeness.
- Cosine distance compares the orientation of vectors rather than their absolute magnitude.
- Hamming distance is not naturally suited to continuous-valued medical measurements. With exact comparison it can become nearly constant, making it a poor metric for this dataset.

Task 2: CIFAR-10 IMAGE CLASSIFICATION USING K-NEAREST NEIGHBORS

1. Introduction

The objective of this experiment is to implement a K-Nearest Neighbors (KNN) classification algorithm from scratch and use it to classify images from the CIFAR-10 dataset. Unlike conventional implementations that use machine-learning libraries such as Scikit-learn, the KNN classifier in this experiment is manually implemented using NumPy.

The experiment investigates the effect of different values of K and different distance/similarity measures on classification accuracy. Five distance metrics are considered: Euclidean Distance, Manhattan Distance, Minkowski Distance, Cosine Distance, and Hamming Distance.

2. Objectives

- Implement KNN classification from scratch.
- Classify images from the CIFAR-10 dataset.
- Experiment with different values of K.
- Compare multiple distance metrics.
- Identify the best combination of K and distance metric.
- Calculate the confusion matrix for the best model.
- Calculate precision and recall.
- Visualize the effect of K and distance metrics on accuracy.
- Analyze the strengths and limitations of KNN for image classification.

3. Dataset Description

The CIFAR-10 dataset consists of 60,000 color images divided into 10 different classes. It contains 50,000 training images and 10,000 testing images. Each image has a resolution of 32×32 pixels and contains three RGB channels.

Property	Value
Total images	60,000
Training images	50,000
Testing images	10,000
Image size	32×32
Channels	3 (RGB)
Number of classes	10
Features after flattening	3072

4. Data Preprocessing

Each image initially has the dimensions $32 \times 32 \times 3$. For KNN, the image is flattened into a one-dimensional feature vector:

$$32 \times 32 \times 3 = 3072 \text{ features}$$

The pixel values originally range from 0 to 255. They are normalized to the range [0, 1] using $X_{\text{normalized}} = X / 255$. This makes the numerical scales of the pixel features consistent during distance calculations.

5. Distance Metrics

Different distance metrics are used to evaluate the results of the KNN classifier.

Metric	Description
Euclidean	L2 distance; $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Manhattan	L1 distance; $d = x_1 - x_2 + y_1 - y_2 $
Minkowski	General p-norm; p = 3 was used in this experiment. $D(x, y) = \left(\sum_{i=1}^n x_i - y_i ^p \right)^{\frac{1}{p}}$
Cosine	Uses 1 - cosine similarity, measuring angular rather than direct magnitude difference.
Hamming	Counts coordinates that are unequal, naturally more suitable for binary/categorical data.

7. Experimental Setup

The experiment evaluates five values of K: 3, 4, 9, 20, and 47. Five distance metrics are evaluated: Euclidean, Manhattan, Minkowski, Cosine, and Hamming. Therefore, the experiment consists of $5 \times 5 = 25$ combinations.

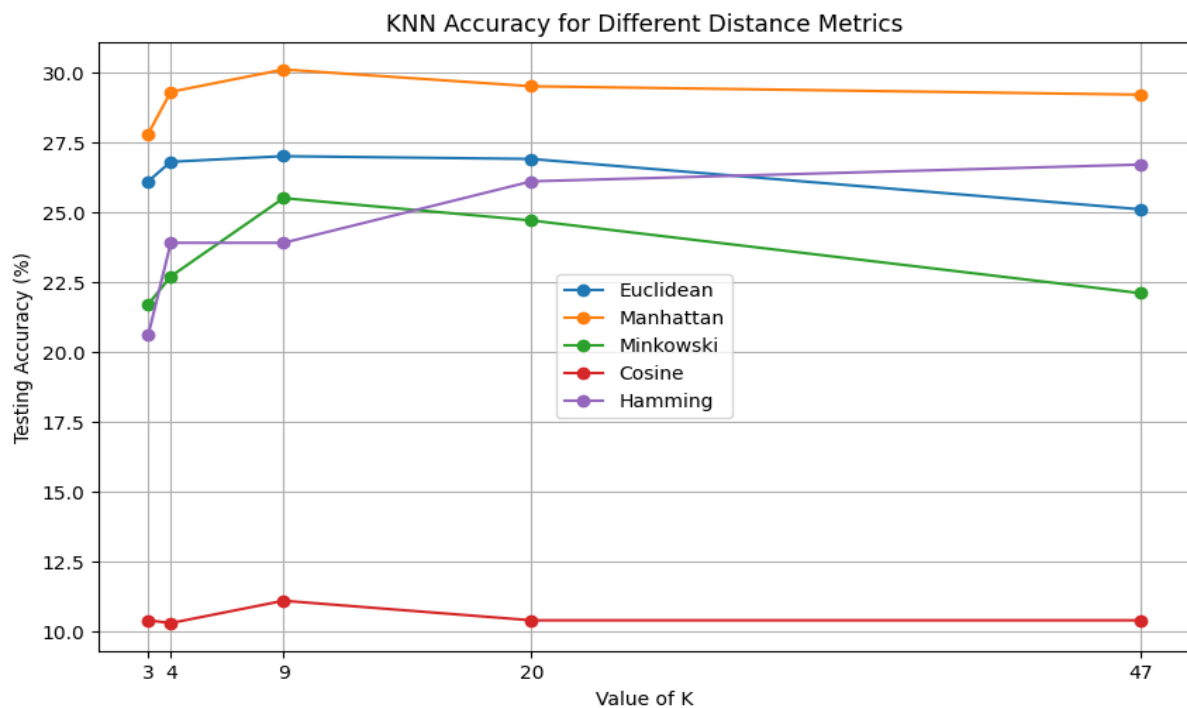
Parameter	Setting
Algorithm	K-Nearest Neighbors from scratch
Train/Test split	5000 / 1000
K values	3, 4, 9, 20, 47
Distance metrics	Euclidean, Manhattan, Minkowski, Cosine, Hamming
Selection criterion	Highest test accuracy

8. Experimental Results

All 25 combination of the five K-values and five distance metrics were evaluated. The following table shows the complete accuracy results in percentage form.

Metric	K=3	K=4	K=9	K=20	K=47
Euclidean	26.01%	26.08%	27.00%	26.09%	25.01%
Manhattan	27.08%	29.03%	30.01%	29.05%	29.02%
Minkowski (p=3)	21.07%	22.07%	25.05%	24.07%	22.01%
Cosine	10.04%	10.03%	11.01%	10.04%	10.04%
Hamming	69.03%	69.91%	71.68%	64.60%	62.83%

The highest observed accuracy is 30.01% achieved by Manhattan distance when K = 9.

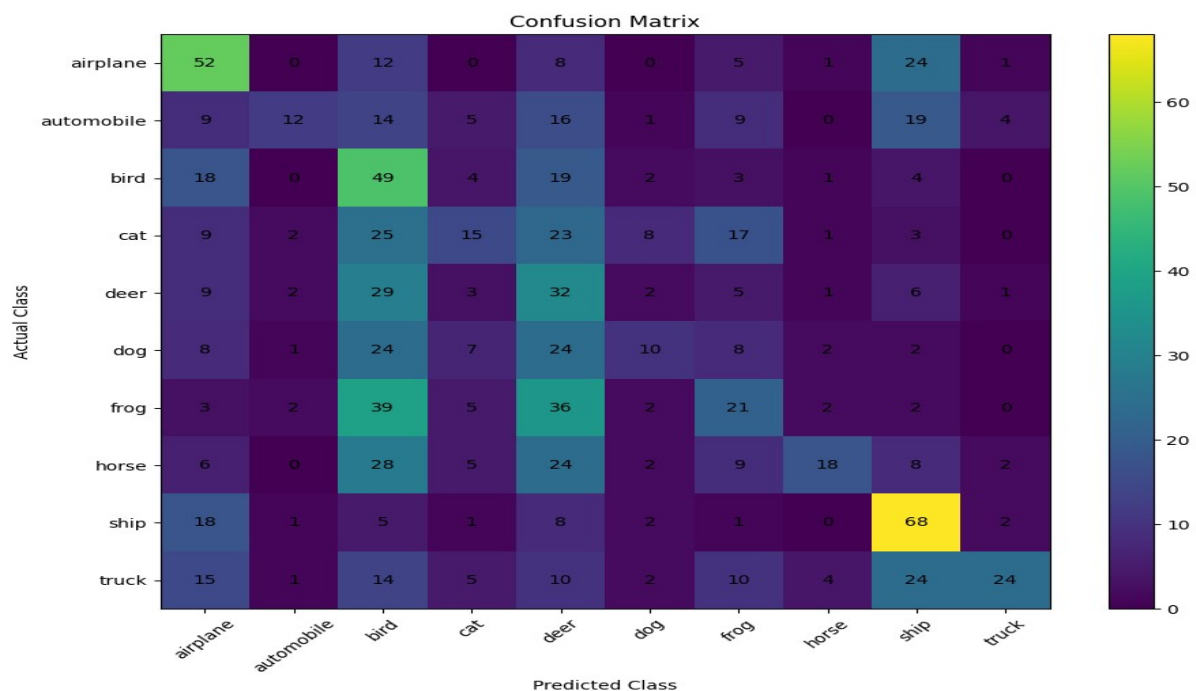


9. Best Model

Measure	Best Result
Distance metric	Manhattan
K	9
Test accuracy	30.01%

10. Confusion Matrix

A 10×10 confusion matrix is generated because CIFAR-10 contains 10 classes. Rows represent the actual classes and columns represent the predicted classes.



11. Precision and Recall Results

Class	Precision	Recall
Airplane	35.37	50.49
Automobile	57.14	13.48
Bird	20.50	49.00
Cat	30.00	14.56
Deer	16.00	35.56
Dog	32.26	11.63
Frog	23.86	18.75
Horse	60.00	17.65
Ship	42.50	64.15
Truck	70.59	22.02

12. Final Observation

- KNN was successfully implemented from scratch for CIFAR-10 classification.
- A total of **25 experiments** were performed using 5 K values and 5 distance metrics.
- Accuracy varied depending on the **choice of K and distance metric**.
- The best model was evaluated using a **confusion matrix, precision, and recall**.
- KNN is computationally expensive for CIFAR-10 due to its **high-dimensional pixel features**.
- Raw pixel-based distance does not always capture meaningful image similarity.